

Indefinite Integrals Involving Logarithmic Functions

17. Find indefinite integrals for each of the following:

(g) $\frac{2x-3}{2x^2-6x+1}$

(h) $\frac{ax+b}{ax^2+2bx+c}$

(i) $\frac{4-2x}{4x-x^2}$

(s) $\frac{e^x - e^{-x}}{e^x + e^{-x}}$

(t) $\frac{1}{e^{-2x} + 1}$

(u) $\frac{3 \sin x}{6 + 5 \cos x}$

(v) $\frac{\sec^2 2x}{1 - 3 \tan 2x}$

(g) $\frac{1}{2} \int \frac{2(2x-3)}{2x^2-6x+1} dx = \frac{1}{2} \ln |2x^2-6x+1| + C$

(h) $\frac{1}{2} \int \frac{2(ax+b)}{ax^2+2bx+c} dx = \frac{1}{2} \ln |ax^2+2bx+c| + C$

(i) $\int \frac{4-2x}{4x-x^2} dx = \ln |4x-x^2| + C$

(s) $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \ln (e^x + e^{-x}) + C$

(t) $\int \frac{1}{e^{-2x} + 1} dx = \int \frac{1}{\frac{1}{e^{2x}} + 1} dx = \frac{1}{2} \int \frac{2e^{2x}}{1+e^{2x}} dx$
 $= \frac{1}{2} \ln (1+e^{2x}) + C$

(u) $\int \frac{3 \sin x}{6+5 \cos x} dx$

$= \frac{3}{-5} \int \frac{-5 \sin x}{6+5 \cos x} dx$

$= -\frac{3}{5} \ln |6+5 \cos x| + C$

(v) $\frac{1}{-6} \int \frac{-6 \sec^2 2x}{1-3 \tan 2x} dx$

$= -\frac{1}{6} \ln |1-3 \tan 2x| + C$

18. Show that $\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$. Hence find $\int \frac{dx}{(x+1)(x+2)}$.

$$\begin{aligned} \int \frac{dx}{(x+1)(x+2)} &= \int \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx \\ &= \ln |x+1| - \ln |x+2| + C \\ &= \ln \left| \frac{x+1}{x+2} \right| + C \end{aligned}$$

Indefinite Integrals Involving Logarithmic Functions continued

19. (a) Show that $\frac{x^2+1}{x+1} = x-1 + \frac{2}{x+1}$. PHS = $x-1 + \frac{2}{x+1} = \frac{x^2-1+2}{x+1} = \frac{x^2+1}{x+1} = \underline{\underline{CHS}}$

(b) Find the equation of the curve for which $\frac{dy}{dx} = \frac{x^2+1}{x+1}$ if the curve passes through the point (0,1).

$$\frac{dy}{dx} = \frac{x^2+1}{x+1}$$

$$y = \int \frac{x^2+1}{x+1} dx$$

$$y = \int \left(x-1 + \frac{2}{x+1} \right) dx$$

$$y = \frac{x^2}{2} - x + 2 \ln|x+1| + C$$

Substitute (0,1)

$$1 = \frac{0^2}{2} - 0 + 2 \ln|0+1| + C$$

$$1 = C$$

∴ equation of the curve is

$$y = \frac{x^2}{2} - x + 2 \ln|x+1| + 1,$$

20. Find the following indefinite integrals:

(a) $\int \frac{4}{x} dx$

(b) $\int \frac{4}{x^5} dx$

(c) $\int \frac{dx}{2x-8}$

(d) $\int \frac{dx}{(2x-8)^2}$

(e) $\int \frac{x}{3x^2+7} dx$

(f) $\int \frac{x}{(3x^2+7)^3} dx$

(a) $\int \frac{4}{x} dx = 4 \ln|x| + C$

(b) $\int \frac{4}{x^5} dx = \frac{4}{6} \int \frac{1}{x^5} dx$

(b) $\int \frac{4}{x^5} dx = 4 \int x^{-5} dx$

$$= 4 \left(\frac{x^{-4}}{-4} \right) + C$$

$$= -\frac{1}{x^4} + C$$

$$= \frac{1}{6} \ln|3x^2+7| + C$$

(c) $\int \frac{dx}{2x-8} = \frac{1}{8} \int \frac{dx}{\frac{1}{2}x-1}$

$$= \frac{1}{6} \frac{(3x^2+7)^{-2}}{-2} + C$$

(c) $\int \frac{dx}{2x-8} = \frac{1}{2} \ln|2x-8| + C$

$$= \frac{1}{-12(3x^2+7)^2} + C$$

(d) $\int (2x-8)^{-2} dx = \frac{(2x-8)^{-1}}{(-1) \cdot 2} + C$

$$= -\frac{1}{2(2x-8)} + C$$

Indefinite Integrals Involving Logarithmic Functions continued

21. Evaluate each of the following definite integrals:

(a) $\int_1^e \frac{(\ln x)^2}{x} dx$

(b) $\int_0^{\frac{\pi}{3}} \tan x dx$

(c) $\int_1^e \frac{(1 + \ln x)^2}{2x} dx$

(d) $\int_e^{e^2} \frac{dw}{w \ln w}$

(e) $\int_1^3 \frac{y dy}{\sqrt{y^2 + 7}}$

(f) $\int_0^2 \frac{x}{(x^2 + 1)^3} dx$

(a) $\int_1^e \frac{1}{x} (\ln x)^2 dx = \left[\frac{(\ln x)^3}{3} \right]_1^e$
 $= \frac{1}{3} [1 - 0] = \frac{1}{3}$

(c) $\frac{1}{2} \int_1^e \frac{1}{x} \cdot (1 + \ln x)^2 dx$
 $= \frac{1}{2} \left[\frac{(1 + \ln x)^3}{3} \right]_1^e$
 $= \frac{1}{6} [8 - 1] = \frac{7}{6}$

(e) $\frac{1}{2} \int_1^3 2y \cdot (y^2 + 7)^{-\frac{1}{2}} dy$
 $= \frac{1}{2} \left[\frac{(y^2 + 7)^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^3 = \sqrt{16} - \sqrt{8}$
 $= 4 - 2\sqrt{2}$

(b) $\int_0^{\frac{\pi}{3}} \tan x dx = - \int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos x} dx$
 $= - \left[\ln |\cos x| \right]_0^{\frac{\pi}{3}}$
 $= - \left[\ln \frac{1}{2} - \ln 1 \right]$
 $= - \ln \left(\frac{1}{2} \right) = \ln 2$

(d) $\int_e^{e^2} \frac{1}{w \ln w} dw$
 $= \left[\ln |\ln w| \right]_e^{e^2}$
 $= \ln 2 - 0 = \ln 2$

(f) $\frac{1}{2} \int_0^2 2x (x^2 + 1)^{-3} dx$
 $= \frac{1}{2} \left[\frac{(x^2 + 1)^{-2}}{-2} \right]_0^2$
 $= \frac{1}{4} \left[\frac{1}{25} - 1 \right]$
 $= \frac{-1}{4} \left[\frac{-24}{25} \right] = \frac{6}{25}$

Applications of Calculus Involving Logarithmic Functions

22. Find the equation of the tangent and the normal to the curve $y = \frac{\ln x}{x}$ at the point where it crosses the x -axis. $\rightarrow y=0$

$y = \frac{\ln x}{x}$

$\frac{\ln x}{x} = 0 \Rightarrow \ln x = 0$
 $\Rightarrow x = e^0$
 $x = 1 \rightarrow (1, 0)$

$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$

$y' = \frac{1 - \ln x}{x^2}$

at $x=1$, $m_T = \frac{1 - \ln(1)}{1^2} = \frac{1 - 0}{1} = 1$

$m_N = \frac{-1}{m_T} = \frac{-1}{1} = -1$

Equation of the tangent at $(1, 0)$:

$y - 0 = 1(x - 1)$

$y = x - 1$

Equation of the normal:

$y - 0 = -1(x - 1)$

$y = -x + 1$

Applications of Calculus Involving Logarithmic Functions continued

23. The total cost \$C\$ of producing \$x\$ textbooks is given by the function
 $C(x) = 800 + 100x - 100 \ln x$ for $x \geq 1$.

- (a) Find the function $A(x)$, an expression in x for the average cost \$A\$ of producing 1 textbook.
- (b) Find the minimum average cost per book.

$$\textcircled{a} \quad A(x) = \frac{800 + 100x - 100 \ln x}{x}$$

$$A(x) = 100 + \frac{800}{x} - \frac{100 \ln x}{x}, \quad x \geq 1$$

$$\textcircled{b} \quad A'(x) = 0 - \frac{800}{x^2} - 100 \left(\frac{x^{-\frac{1}{x}} - \ln x \cdot x^{-1}}{x^2} \right)$$

$$A'(x) = -\frac{800}{x^2} - 100 \left(\frac{1 - \ln x}{x^2} \right)$$

S.P.:

$$A'(x) = 0$$

$$-\frac{800}{x^2} = \frac{100(1 - \ln x)}{x^2}$$

$$-8 = 1 - \ln x$$

$$-8 - 1 = -\ln x$$

$$\ln x = 9 \Rightarrow x = e^9$$

$$A'(x) = -800 \cdot x^{-2} - 100 \left(\frac{1 - \ln x}{x^2} \right)$$

$$A''(x) = -800(-2 \cdot x^{-3}) - 100 \left[\frac{x^2(-\frac{1}{x^2}) - (1 - \ln x) \cdot 2x}{x^4} \right]$$

$$= \frac{1600}{x^3} - 100 \left[\frac{-x - 2x + 2x \ln x}{x^4} \right]$$

$$= \frac{1600}{x^3} - 100 \left(\frac{x(2 \ln x - 3)}{x^4} \right)$$

$$= \frac{1600 - 200 \ln x + 300}{x^3}$$

$$A''(x) = \frac{1900 - 200 \ln x}{x^3}$$

$$\text{at } x = e^9, \quad A''(e^9) = \frac{1900 - 200 \ln(e^9)}{(e^9)^3} = \frac{100}{e^{27}} > 0$$

Minimum average cost at $x = e^9$

$$\therefore x = e^9$$

$$A(e^9) = 100 + \frac{800}{e^9} - \frac{100 \ln(e^9)}{e^9}$$

$$= 100 - \frac{100}{e^9}$$

$$\approx \$99.99$$

Applications of Calculus Involving Logarithmic Functions continued

24. Consider the function $y = \frac{\ln x}{x}$.

- (a) Find the domain. $x > 0$
- (b) Find the x -intercept. $y = 0 \Rightarrow \frac{\ln x}{x} = 0 \Rightarrow \ln x = 0 \Rightarrow x = e^0 = 1$
- (c) Find the values of x for which the graph of the function is below the x -axis.
- (d) Find the coordinates of the stationary point and determine its nature.
- (e) Find the equation of the vertical asymptote of the function $y = \frac{\ln x}{x}$ and,

given that $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$, write down the equation of the horizontal asymptote.

- (f) Show that $\left(e^{\frac{3}{2}}, \frac{3e^{-\frac{3}{2}}}{2}\right)$ is a point of inflection.

- (g) Sketch the graph of the function $y = \frac{\ln x}{x}$.

(c) $y < 0$

$$\frac{\ln x}{x} < 0$$

$$\Rightarrow 0 < x < 1$$

(d) $y = \frac{\ln x}{x}$

$$y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$$

$$y' = \frac{1 - \ln x}{x^2}$$

S.P.: $y' = 0$

$$\frac{1 - \ln x}{x^2} = 0$$

$$1 - \ln x = 0$$

$$1 = \ln x$$

$$\Rightarrow x = e^1 = e, y = \frac{\ln e}{e} = \frac{1}{e}$$

$\therefore (e, \frac{1}{e})$ is the stationary point

$$y' = \frac{1 - \ln x}{x^2}$$

$$y'' = \frac{x^2(0 - \frac{1}{x}) - (1 - \ln x) \cdot 2x}{x^4}$$

$$= \frac{-x - 2x + 2x \ln x}{x^4}$$

$$= \frac{x(2 \ln x - 3)}{x^4} = \frac{2 \ln x - 3}{x^3}$$

$$\therefore y'(x) = \frac{2 \ln x - 3}{x^3}$$

$$x = e, y'(e) = \frac{2 \ln e - 3}{e^3} = \frac{-1}{e^3} < 0$$

Maximum T.P at $(e, \frac{1}{e})$

(e) V.A: $x = 0$ ($y \rightarrow \pm \infty$)
H.A: $y = 0$ ($x \rightarrow \pm \infty$)

(f) $y''(x) = 0$

$$\frac{2 \ln x - 3}{x^3} = 0$$

$$2 \ln x = 3$$

$$\ln x = \frac{3}{2}$$

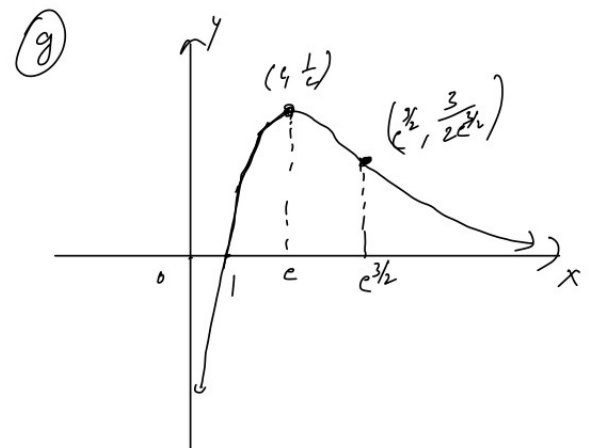
$$x = e^{\frac{3}{2}}, y = \frac{\ln e^{\frac{3}{2}}}{e^{\frac{3}{2}}} = \frac{3}{2e^{\frac{3}{2}}}$$

$(e^{\frac{3}{2}}, \frac{3}{2e^{\frac{3}{2}}})$ is the possible P.O.I

x	1	$e^{\frac{3}{2}}$	∞
y''	-	0	+

concavity changes

$\therefore (e^{\frac{3}{2}}, \frac{3}{2e^{\frac{3}{2}}})$ is the P.O.I



Applications of Calculus Involving Logarithmic Functions continued

25. Find the area of the region bounded by the line $x + y = 5$ and the curve $xy = 4$.

$$y = 5 - x \quad \begin{matrix} 5 - x = \frac{4}{x} \\ y = \frac{4}{x} \end{matrix}$$

$$5x - x^2 = 4$$

$$-x^2 + 5x - 4 = 0$$

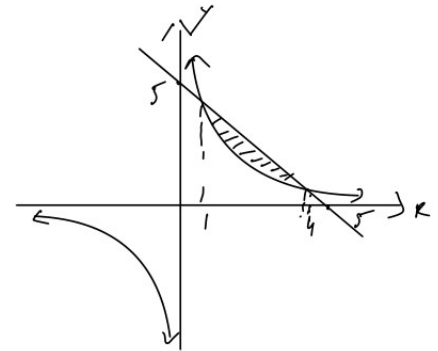
$$(x-1)(x-4) = 0$$

$$x = 1 \text{ or } x = 4$$

$$y = 4 \quad y = 1$$

$$(1, 4) \quad (4, 1)$$

$$\begin{aligned} \text{Area, } A &= \int_1^4 \left(5 - x - \frac{4}{x} \right) dx \\ &= \left[5x - \frac{x^2}{2} - 4 \ln|x| \right]_1^4 \\ &= \left(20 - 8 - 4 \ln 4 \right) - \left(5 - \frac{1}{2} - 4 \ln 1 \right) \\ &= 12 - 4 \ln 4 - 4 \frac{1}{2} \\ A &= \left(7 \frac{1}{2} - 4 \ln 4 \right) \text{ units}^2 \quad \checkmark \end{aligned}$$



26. The region bounded by the curve $y = \sqrt{\frac{x}{x^2 + 1}}$, the x -axis and the line $x = 2$ is rotated about the x -axis. Find the volume of the solid of revolution.
[Note that the curve passes through the origin and lies in the first quadrant.]

$$\begin{aligned} V &= \pi \int_0^2 \left(\sqrt{\frac{x}{x^2 + 1}} \right)^2 dx \\ &= \frac{\pi}{2} \int_0^2 \frac{2x}{x^2 + 1} dx \\ &= \frac{\pi}{2} \left[\ln(x^2 + 1) \right]_0^2 \\ &= \frac{\pi}{2} [\ln 5 - \ln 1] \\ &= \frac{\pi}{2} \ln 5 \text{ units}^3 \end{aligned}$$

Differentiation of Trigonometric Functions

27. Differentiate with respect to x :

(g) $\tan(4x^2 - 1)$

(h) $\cos x^\circ$

(i) $x + 3 \sin x - \cos 2x$

(j) $4 \sin(3x - 5)$

(k) $x^3 \cos 2x$

(l) $\sin 2x \tan 3x$

(m) $\frac{4x}{\cos x}$

(n) $\frac{x+1}{\sin x}$

(o) $(\sin x - \cos x)^2$

(g) $\left[\tan(4x^2 - 1) \right]' = 8x \cdot \sec^2(4x^2 - 1)$

(h) $\left[\cos x^\circ \right]' = \left[\cos \frac{\pi x}{180} \right]' = -\frac{\pi}{180} \sin x^\circ$

(i) $\left[x + 3 \sin x - \cos 2x \right]' = 1 + 3 \cos x + 2 \sin 2x$

(j) $\left[4 \sin(3x - 5) \right]' = 12 \cos(3x - 5)$

(k) $\left[x^3 \cos 2x \right]' = -2x^3 \sin 2x + 3x^2 \cos 2x = x^2 (3 \cos 2x - 2x \sin 2x)$

(l) $\left[\sin 2x \cdot \tan 3x \right]' = \sin 2x (3 \sec^2 3x) + \tan 3x \cdot (2 \cos 2x)$

(m) $\left[\frac{4x}{\cos x} \right]' = \frac{\cos x \cdot 4 - 4x(-\sin x)}{\cos^2 x} = \frac{4 \cos x + 4x \sin x}{\cos^2 x}$

(n) $\left[\frac{x+1}{\sin x} \right]' = \frac{\sin x \cdot (1) - (x+1) \cdot \cos x}{\sin^2 x} = \frac{\sin x - (x+1) \cos x}{\sin^2 x}$

(o) $\left[(\sin x - \cos x)^2 \right]' = \left(\sin^2 x - 2 \sin x \cos x + \cos^2 x \right)'$

$= (1 - \sin 2x)'$

$= 0 - 2 \cos 2x$

$= -2 \cos 2x$

Differentiation of Trigonometric Functions continued

28. Find the second derivative of:

(a) $\cos x + \sin x$

$$\begin{aligned} y &= \cos x + \sin x \\ y' &= -\sin x + \cos x \\ y'' &= -\cos x - \sin x \\ y''' &= -(\cos x + \sin x) \end{aligned}$$

(b) $\cos^2 x$

$$\begin{aligned} y &= \cos^2 x \\ y' &= 2\cos x (-\sin x) \\ y' &= -\sin 2x \\ y'' &= -\cos 2x \cdot 2 \\ y'' &= -2\cos 2x \end{aligned}$$

(c) $e^{\sin x}$

$$\begin{aligned} y &= e^{\sin x} \\ y' &= e^{\sin x} \cdot \cos x \\ y'' &= e^{\sin x} (-\sin x) + \cos x \cdot e^{\sin x} \cdot \cos x \\ y'' &= e^{\sin x} (-\sin x + \cos^2 x) \end{aligned}$$

29. Find the equation of the normal to the curve $y = \frac{1}{\pi} \cos\left(\pi x - \frac{\pi}{3}\right)$ at the point where $x = 1$. Give your answer in general form.

$$\begin{aligned} y &= \frac{1}{\pi} \cos\left(\pi x - \frac{\pi}{3}\right) \\ y' &= \frac{1}{\pi} \left[-\sin\left(\pi x - \frac{\pi}{3}\right)\right] \cdot \pi \end{aligned}$$

$$y' = -\sin\left(\pi x - \frac{\pi}{3}\right)$$

$$\begin{aligned} \text{at } x=1, \quad m_T &= -\sin\left(\pi - \frac{\pi}{3}\right) \\ &= -\sin \frac{\pi}{3} \\ &= -\frac{\sqrt{3}}{2} \end{aligned}$$

$$m_N = \frac{-1}{m_T} = \frac{-1}{-\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$$

Point: $x=1$

$$\begin{aligned} y &= \frac{1}{\pi} \cos\left(\pi - \frac{\pi}{3}\right) \\ &= \frac{1}{\pi} (-\cos \frac{\pi}{3}) \\ &= \frac{1}{\pi} \left(-\frac{1}{2}\right) = -\frac{1}{2\pi} \end{aligned}$$

$(1, -\frac{1}{2\pi})$

Equation of Normal:

$$y + \frac{1}{2\pi} = \frac{2}{\sqrt{3}}(x-1)$$

$$y = \frac{2}{\sqrt{3}}x - \frac{2}{\sqrt{3}} - \frac{1}{2\pi}$$

$$y = \frac{4\pi x - 4\pi - \sqrt{3}}{2\sqrt{3}\pi}$$

$$4\pi x - 2\sqrt{3}\pi y - (4\pi + \sqrt{3}) = 0$$

Differentiation of Trigonometric Functions continued

30. Find the turning point on the curve $y = e^{-x} \sin x$ for $0 \leq x \leq 2$ and determine its nature.

$$y' = e^{-x} (\cos x + \sin x (-e^{-x}))$$

S.P. $y' = 0$

$$e^{-x} (\cos x - \sin x) = 0$$

$$\cos x = \sin x$$

$$1 = \tan x$$

$$x = \frac{\pi}{4} \checkmark (0 \leq x \leq 2)$$

$$x = \frac{\pi}{4}, y = e^{-\frac{\pi}{4}} \cdot \left(\frac{1}{\sqrt{2}}\right)$$

$$\left(\frac{\pi}{4}, \frac{e^{-\frac{\pi}{4}}}{\sqrt{2}}\right) \text{ is the T.P.}$$

Nature:

$$y' = e^{-x} (\cos x - \sin x)$$

$$y'' = e^{-x} (-\sin x - \cos x) + (\cos x - \sin x) \cdot (-e^{-x})$$

$$= e^{-x} (-\sin x - \cos x - \cos x + \sin x)$$

$$y'' = -2 \cos x \cdot e^{-x}$$

at $x = \frac{\pi}{4}, y'' = -2 \cos \frac{\pi}{4} \cdot e^{-\frac{\pi}{4}} < 0$

$\therefore y'' < 0$

Maximum T.P. at $x = \frac{\pi}{4} \checkmark$

31. Consider the function $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$.

- (a) Find $f(0)$ and $f(2\pi)$.
- (b) Show that there is a horizontal point of inflection at the point where $x = \pi$.
- (c) Except where $x = \pi$, show that $f'(x) > 0$ for all values of x in the domain.
- (d) Sketch the graph of $y = x + \sin x$ for $0 \leq x \leq 2\pi$ showing the above features.

(a) $f(0) = 0 + \sin 0 = 0$

$$f(2\pi) = 2\pi + \sin 2\pi = 2\pi + 0 = 2\pi$$

$$f'(x) = 1 + \cos x$$

x	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$f'(x)$	+	0	+

(b) $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$

$$f'(x) = 1 + \cos x$$

$$f'(x) = 0$$

$$\cos x = -1$$

$$x = \pi$$

$$f(\pi) = \pi + \sin \pi = \pi + 0 = \pi$$

(π, π) is the turning point

$$f'(x) = 1 + \cos x$$

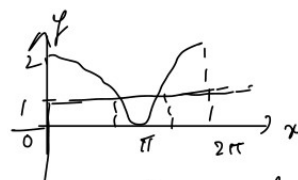
$$f''(x) = 0 - \sin x = -\sin x$$

$$x = \pi, f''(\pi) = -\sin \pi = 0$$

Second derivative test fails

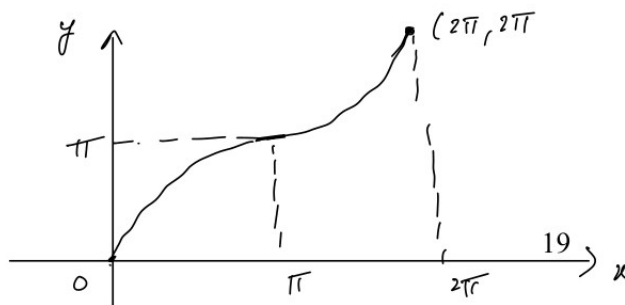
(c) $f'(x) = 1 + \cos x$ ($0 \leq x \leq 2\pi$)

$$1 + \cos x$$



except $x = \pi, f'(x) > 0$ for all x in the domain

(d)



Integration of Basic Trigonometric Functions

32. Find the primitive (indefinite integral) of the following:

(g) $\sin 2x + \cos 3x$

(h) $\cos(2x-3)$

(i) $\sec^2\left(\frac{1}{3}x-2\right)$

(j) $3\sin\left(\frac{\pi}{2}-x\right)$

(k) $\frac{1}{\cos^2 x}$

(l) $\cos\frac{x}{2} + \frac{1}{2}\sin 2x$

(g) $\int (\sin 2x + \cos 3x) dx = -\frac{\cos 2x}{2} + \frac{\sin 3x}{3} + C$

(h) $\int \cos(2x-3) dx = \frac{\sin(2x-3)}{2} + C$

(i) $\int \sec^2\left(\frac{1}{3}x-2\right) dx = 3 \tan\left(\frac{1}{3}x-2\right) + C$

(j) $\int 3 \sin\left(\frac{\pi}{2}-x\right) dx = \int 3 \cos x dx = 3 \sin x + C$

(k) $\int \frac{1}{\cos^2 x} dx = \int \sec^2 x dx = \tan x + C$

(l) $\int \left(\cos\frac{x}{2} + \frac{1}{2} \sin 2x \right) dx = \frac{\sin\frac{x}{2}}{\frac{1}{2}} + \frac{1}{2} \left(\frac{-\cos 2x}{2} \right) + C$
 $= 2 \sin\frac{x}{2} - \frac{1}{4} \cos 2x + C$